

# Electric Circuits



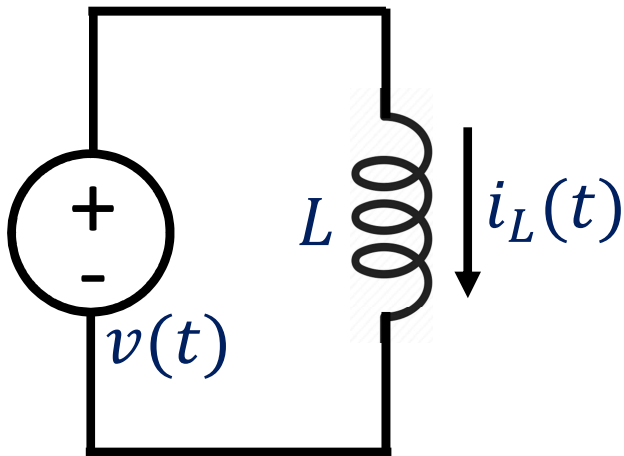
*Series RL and RC Circuits*

# Tentative Lecture Plan

| L                                     | LDay-1      | LDay-2           | Topics                                                                              | T | TDay        |
|---------------------------------------|-------------|------------------|-------------------------------------------------------------------------------------|---|-------------|
| 1                                     | FRI, 04 AUG | MON, 07 AUG      | Basic Definitions, Voltage & Current Sources, KCL & KVL                             | 1 | THU, 10 AUG |
| 2                                     | TUE, 08 AUG | FRI, 11 AUG      | Nodal Analysis: Independent and Dependent Current Sources                           | 2 | SAT, 19 AUG |
| 3                                     | MON, 14 AUG | THU, 17 AUG      | Nodal Analysis: Current and Voltage Sources                                         |   |             |
| 4                                     | FRI, 18 AUG | MON, 21 AUG      | Mesh Analysis: Voltage and Current Sources                                          | 3 | THU, 24 AUG |
| 5                                     | TUE, 22 AUG | FRI, 25 AUG      | Network Theorems: Superposition Theorem                                             | R | SAT, 26 AUG |
| SUN, 27 AUG : Quiz-1 : Lectures 1-5   |             |                  |                                                                                     |   |             |
| 6                                     | MON, 28 AUG | TUE, 29 AUG      | Network Theorems: Thevenin's and Norton's Theorem                                   | 4 | THU, 31 AUG |
| 7                                     | MON, 04 SEP | TUE, 05 SEP      | Network Theorems: Maximum Power Transfer Theorem                                    | 5 | WED, 06 SEP |
| 8                                     | FRI, 08 SEP | SAT (Mo), 09 SEP | Series RL and RC Circuits                                                           |   |             |
| 9                                     | MON, 11 SEP | TUE, 12 SEP      | Series RL, RC, RLC Circuits<br>Response to Constant, Sinusoid and Exponential Input | 6 | THU, 14 SEP |
| 10                                    | FRI, 15 SEP | SAT (Tu), 16 SEP | Concept of Phasor, AC Circuits, AC Power                                            |   |             |
| WED, 20 SEP: MidSem : Lectures 1 – 10 |             |                  |                                                                                     |   |             |

# Circuit Element: Inductor

An inductor is a two terminal device that develops a voltage drop  $v_L(t)$  that consists of only induced voltage due to time-varying magnetic fields. The unit of inductance is Henry and is abbreviated as **H**.



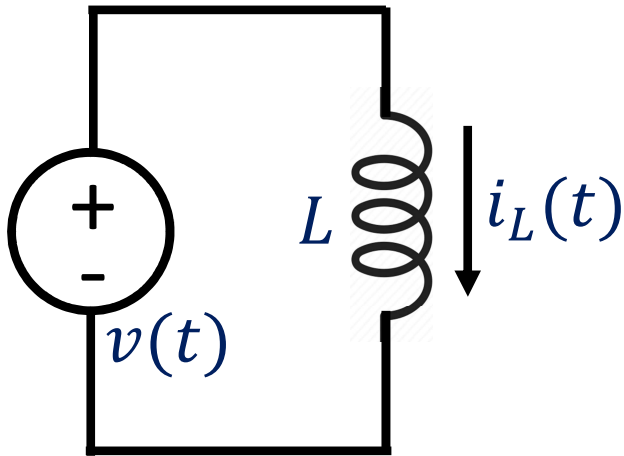
$$v_L(t) = L \frac{di_L(t)}{dt}$$

$$i_L(t) = \frac{1}{L} \int_{-\infty}^t v_L(\tau) d\tau$$

$$i_L(t) = \frac{1}{L} \int_{-\infty}^0 v_L(\tau) d\tau + \frac{1}{L} \int_0^t v_L(\tau) d\tau$$

$$i_L(t) = i_L(0) + \frac{1}{L} \int_0^t v_L(\tau) d\tau$$

# Circuit Element: Inductor



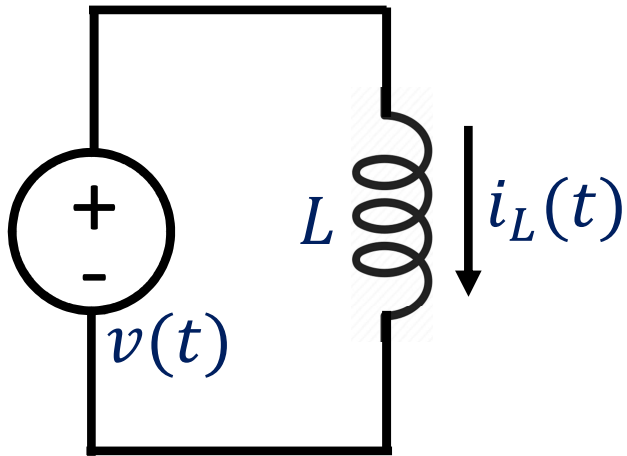
$$v_L(t) = L \frac{di_L(t)}{dt}$$

$$P_L(t) = v_L(t)i_L(t)$$

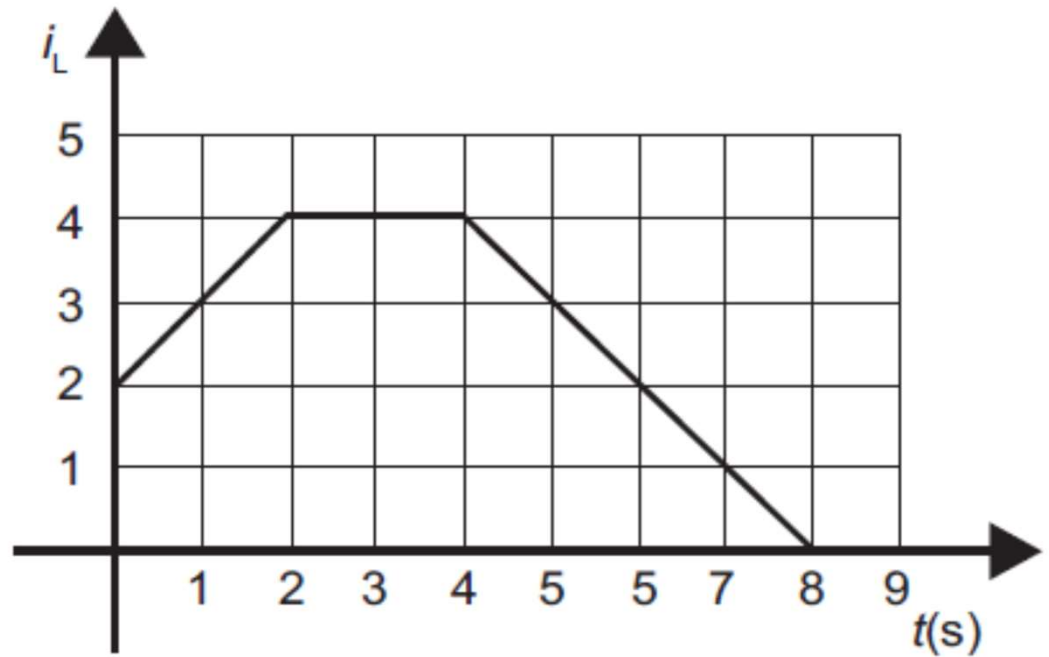
$$\begin{aligned} E_L(t) - E_L(0) &= \int_0^t v_L(\tau)i_L(\tau)d\tau \\ &= \int_0^t i_L(\tau)L \frac{di_L(\tau)}{d\tau} d\tau = L \int_0^t i_L(\tau)d\{i_L(\tau)\} \end{aligned}$$

$$E_L(t) - E_L(0) = \frac{L}{2} [\{i_L(t)\}^2 - \{i_L(0)\}^2]$$

# Circuit Element: Inductor

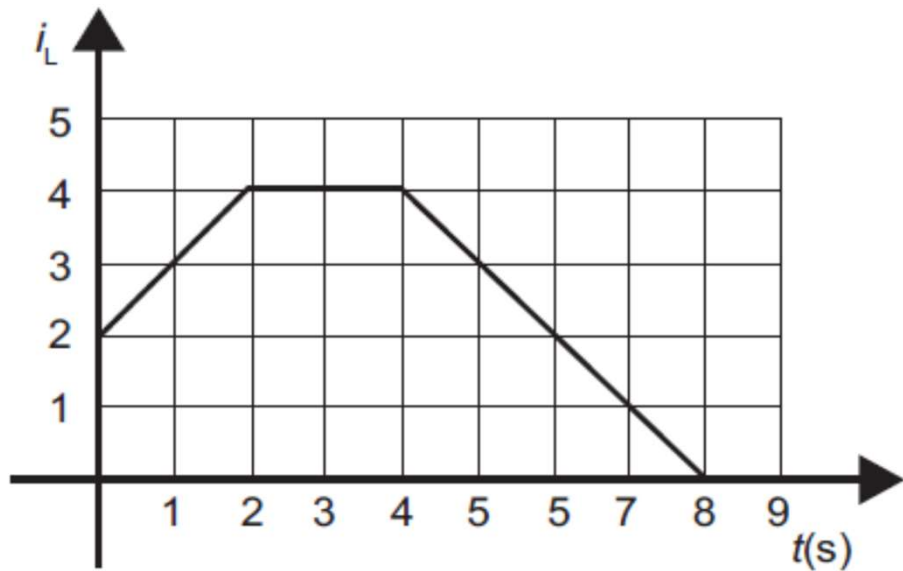


$$L = 2H$$



Plot  $v(t)$ ,  $P_L(t)$  and  $E_L(t)$

# Circuit Element: Inductor



$$i_L(0) = 2$$

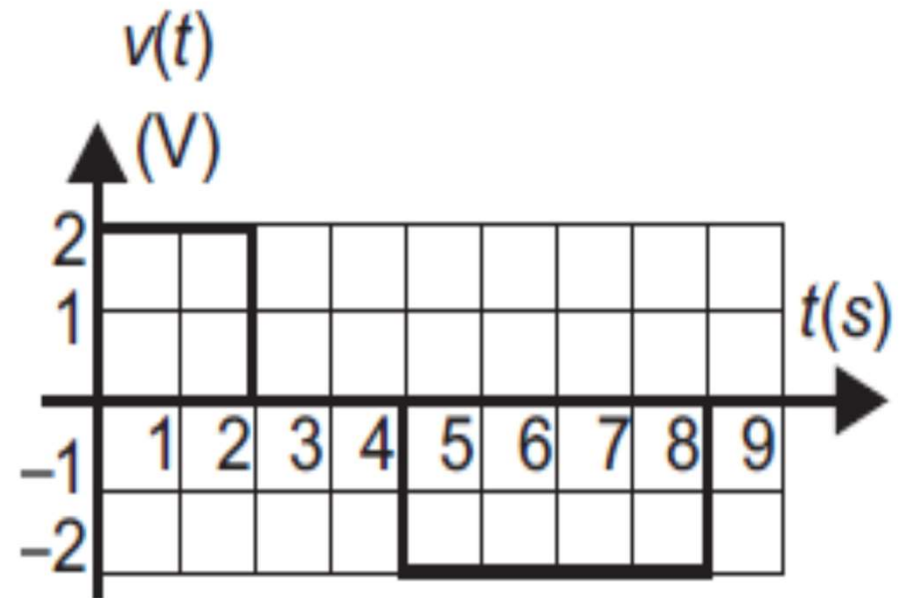
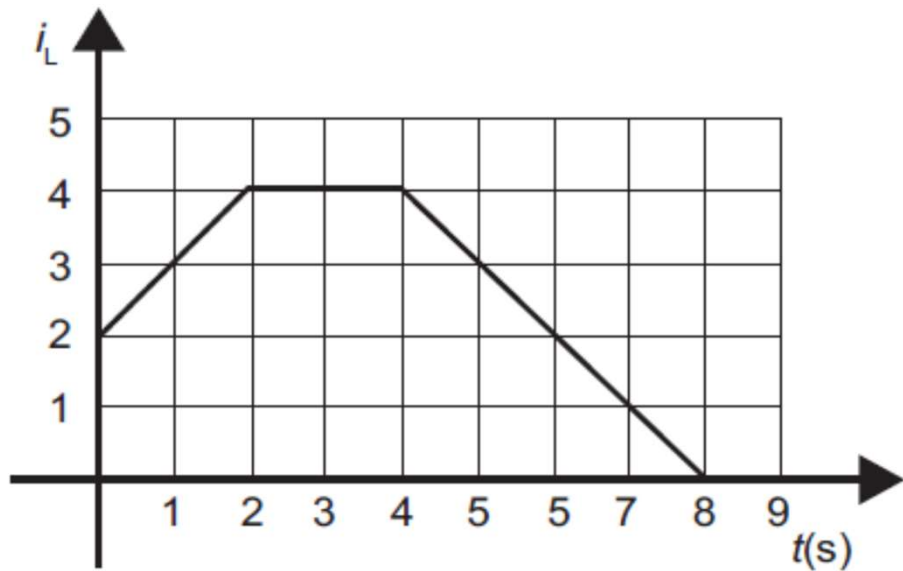
$$(0,2): i_L(t) = t + 2$$

$$(2,4): i_L(t) = 4$$

$$(4,8): i_L(t) = 8 - t$$

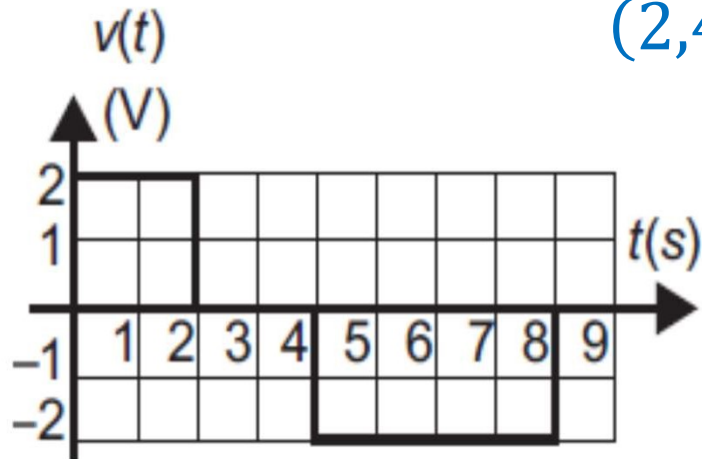
$$(t \geq 8): i_L(t) = 0$$

# Circuit Element: Inductor

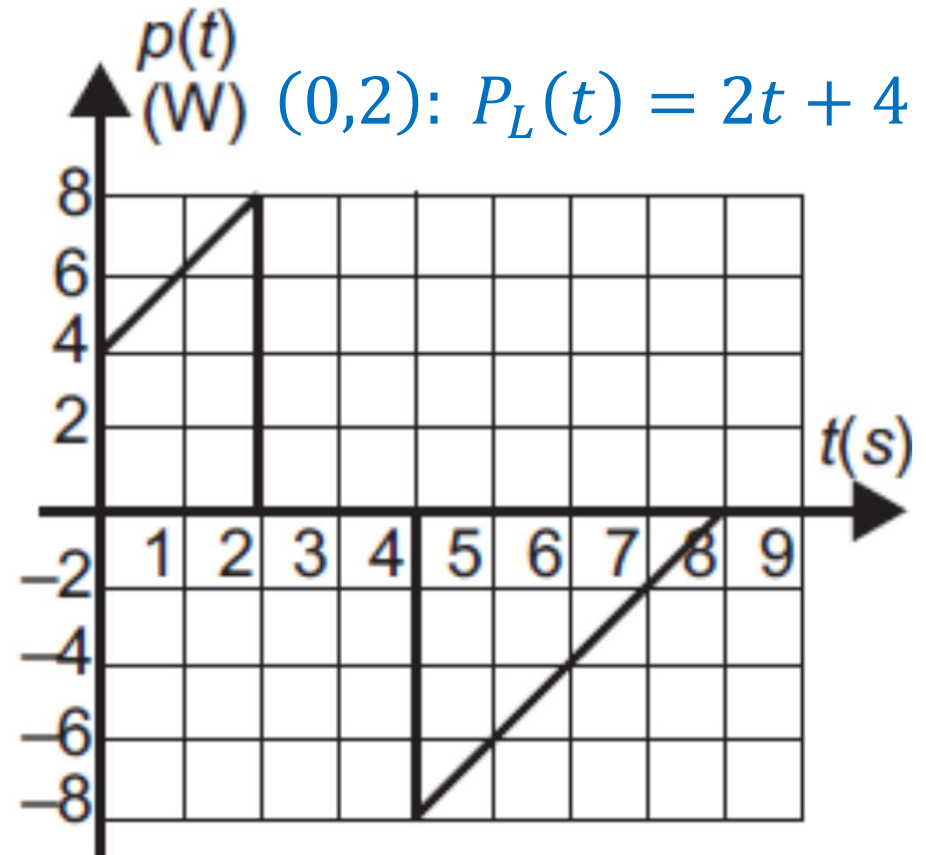


$$v_L(t) = L \frac{di_L(t)}{dt}$$

# Circuit Element: Inductor



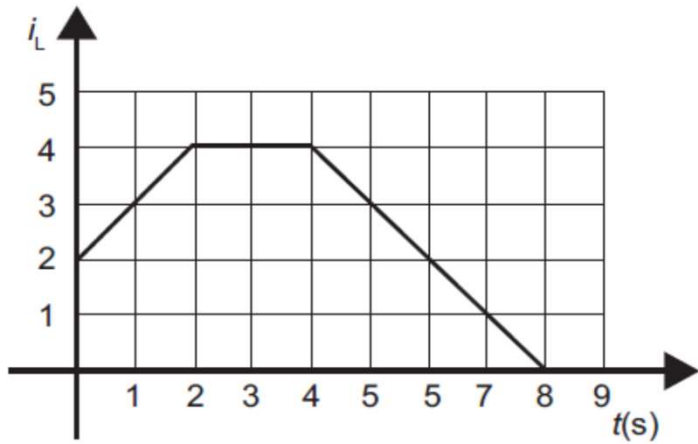
$$(2,4): P_L(t) = 0$$



$$(4,8): P_L(t) = 2t - 16$$



# Circuit Element: Inductor

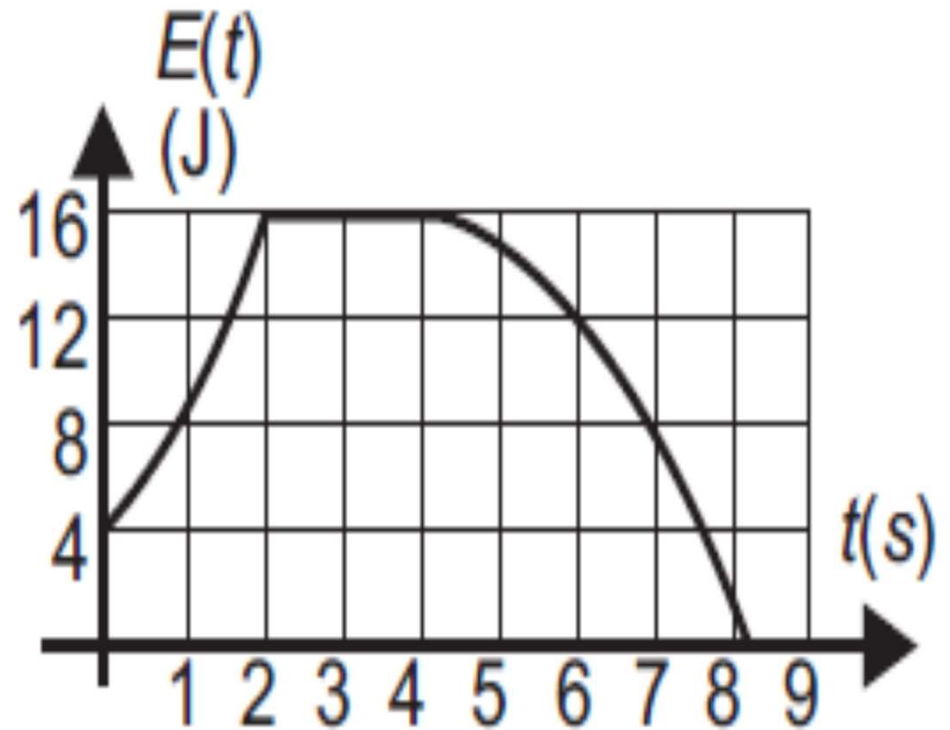


$$E_L(0) = \frac{LI_0^2}{2} = 4J$$

$$(0,2): E_L(t) = t^2 + 4t + 4$$

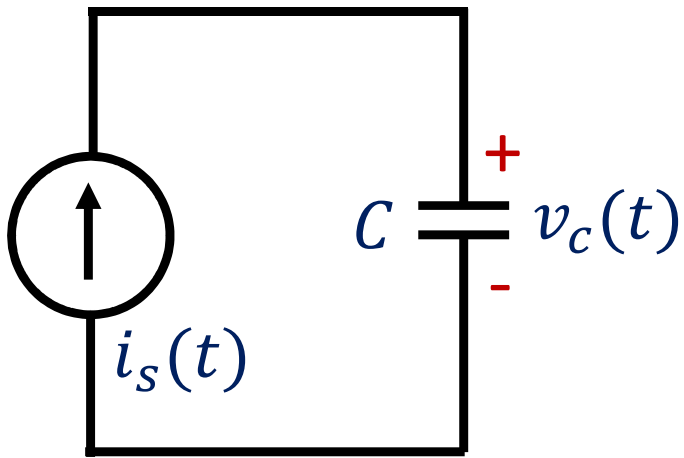
$$(2,4): E_L(t) = 16$$

$$(4,8): E_L(t) = t^2 - 16t + 64$$



# Circuit Element: Capacitor

A capacitor is a two terminal device having a current proportional to the rate of change of voltage across it. The unit of capacitance is Farad and is abbreviated as **F**.



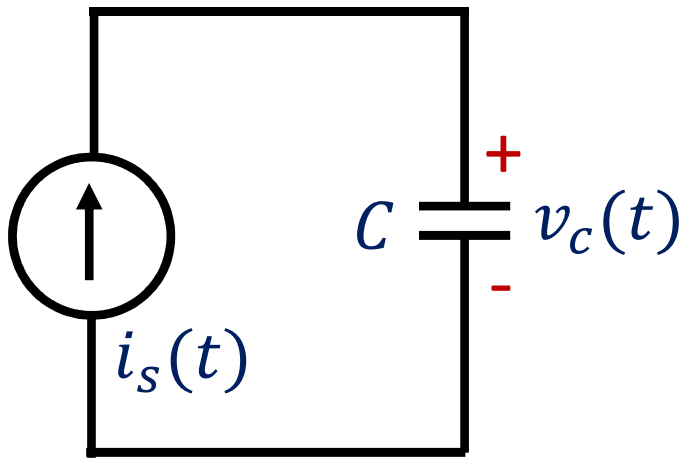
$$i_c(t) = C \frac{dv_c(t)}{dt}$$

$$v_c(t) = \frac{1}{C} \int_{-\infty}^t i_c(\tau) d\tau$$

$$v_c(t) = \frac{1}{C} \int_{-\infty}^0 i_c(\tau) d\tau + \frac{1}{C} \int_0^t i_c(\tau) d\tau$$

$$v_c(t) = v_c(0) + \frac{1}{C} \int_0^t i_c(\tau) d\tau$$

# Circuit Element: Capacitor



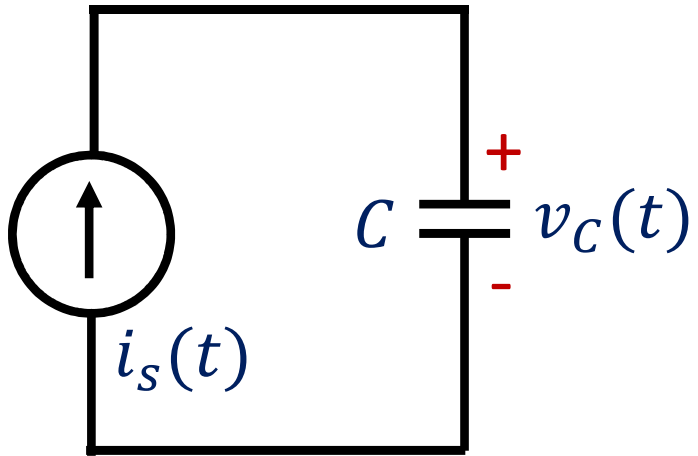
$$i_c(t) = C \frac{dv_c(t)}{dt}$$

$$P_c(t) = v_c(t)i_c(t)$$

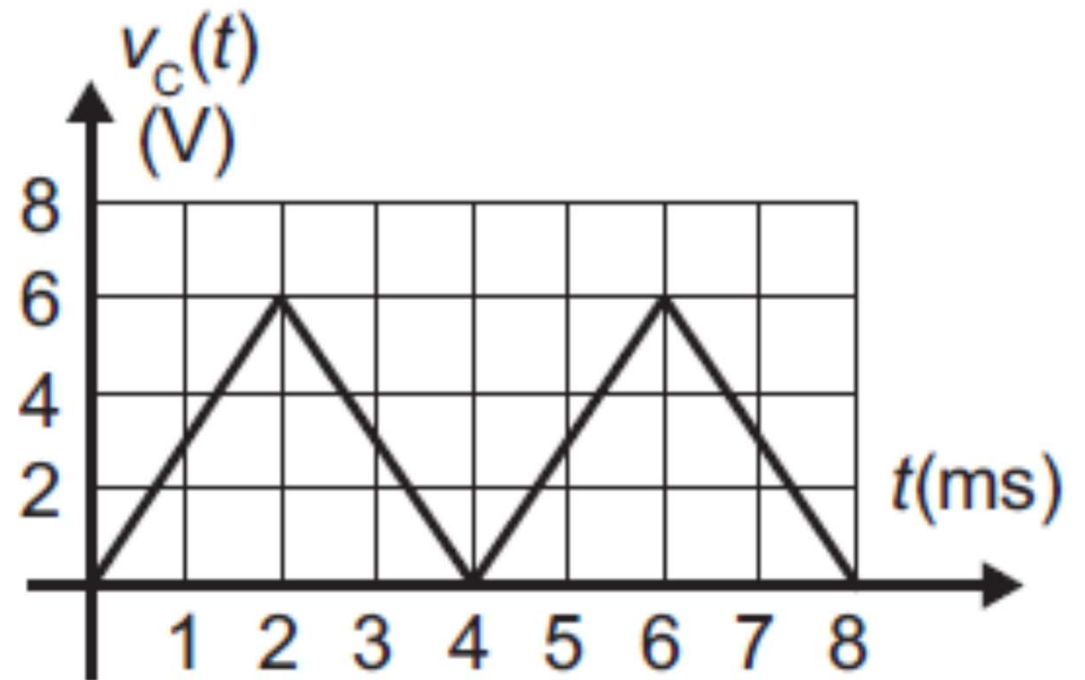
$$\begin{aligned} E_c(t) - E_c(0) &= \int_0^t v_c(\tau) i_c(\tau) d\tau \\ &= \int_0^t v_c(\tau) C \frac{dv_c(\tau)}{d\tau} d\tau = C \int_0^t v_c(\tau) d\{v_c(\tau)\} \end{aligned}$$

$$E_c(t) - E_c(0) = \frac{C}{2} [\{v_c(t)\}^2 - \{v_c(0)\}^2]$$

# Circuit Element: Capacitor

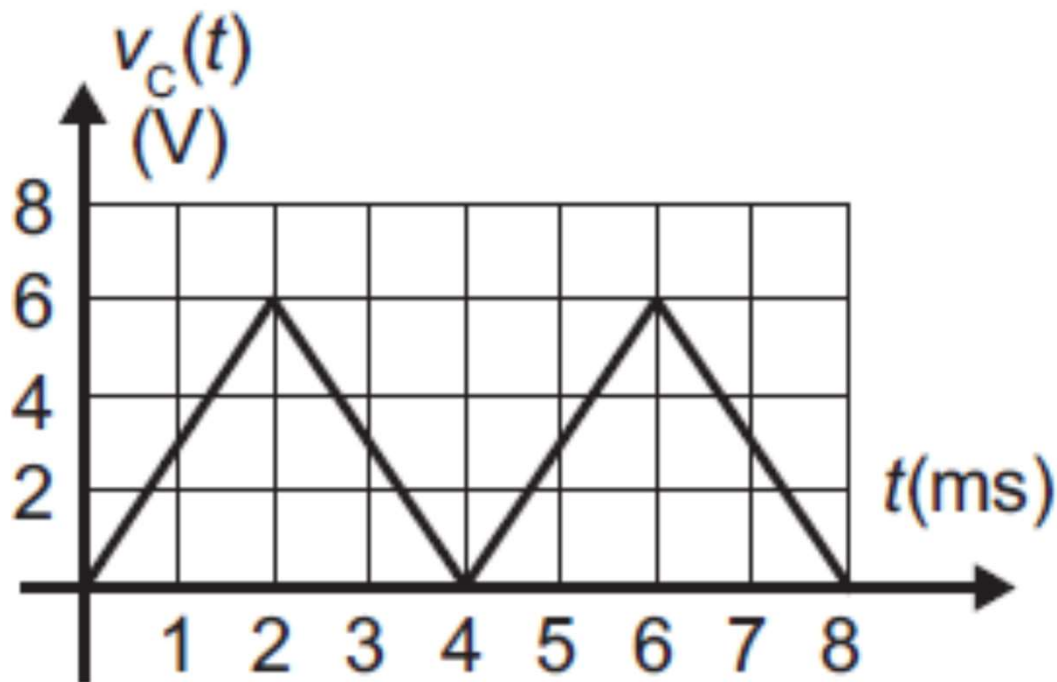


$$C = 1000\mu F$$



Plot  $i(t)$ ,  $P_c(t)$  and  $E_c(t)$

# Circuit Element: Capacitor



$$v_c(0) = 0$$

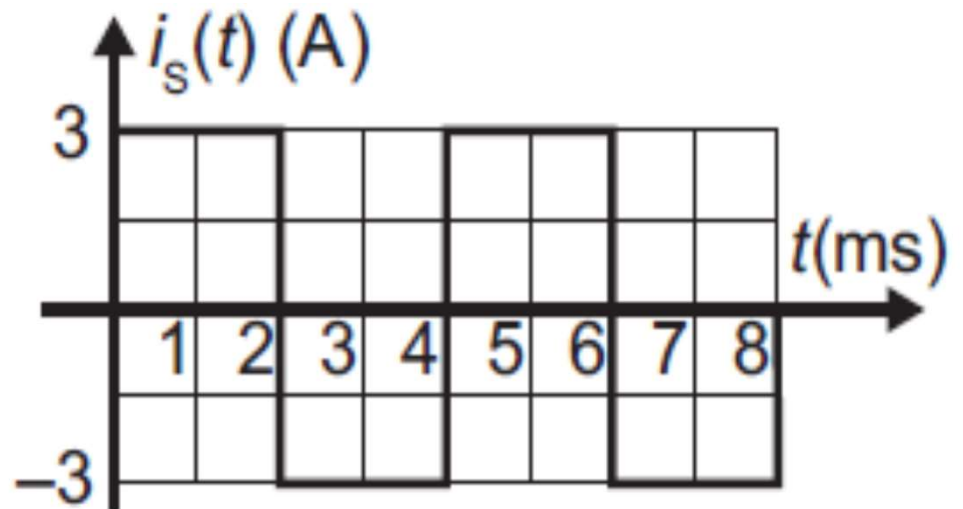
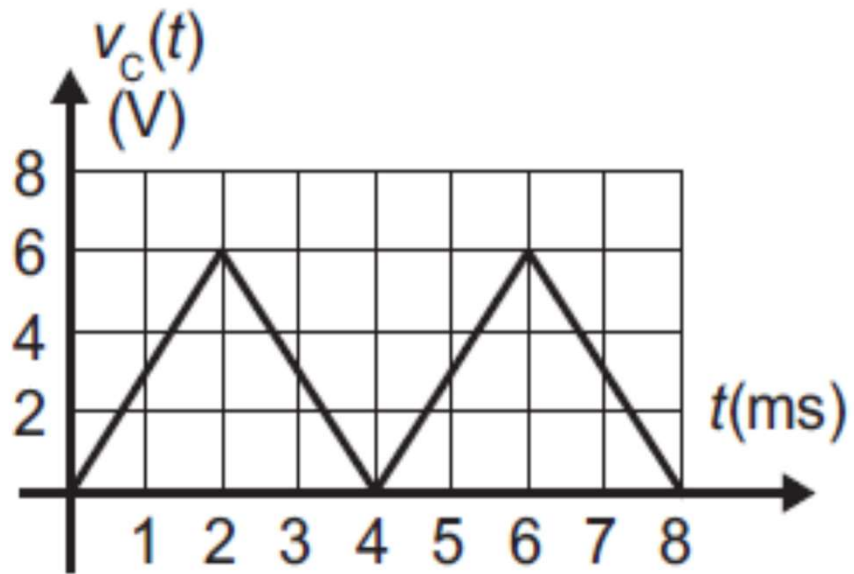
$$(0,2): v_c(t) = 3t$$

$$(2,4): v_c(t) = 12 - 3t$$

$$(4,6): v_c(t) = 3t - 12$$

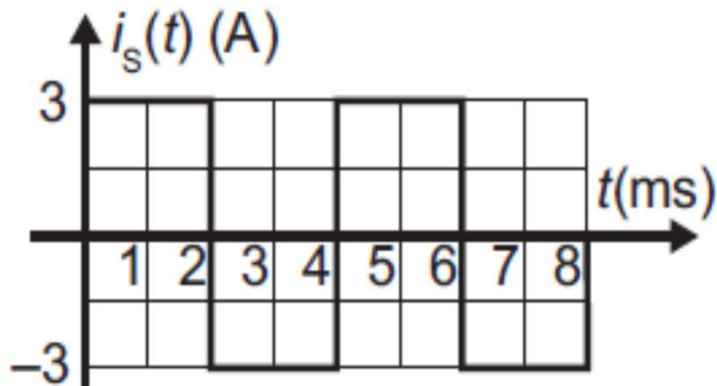
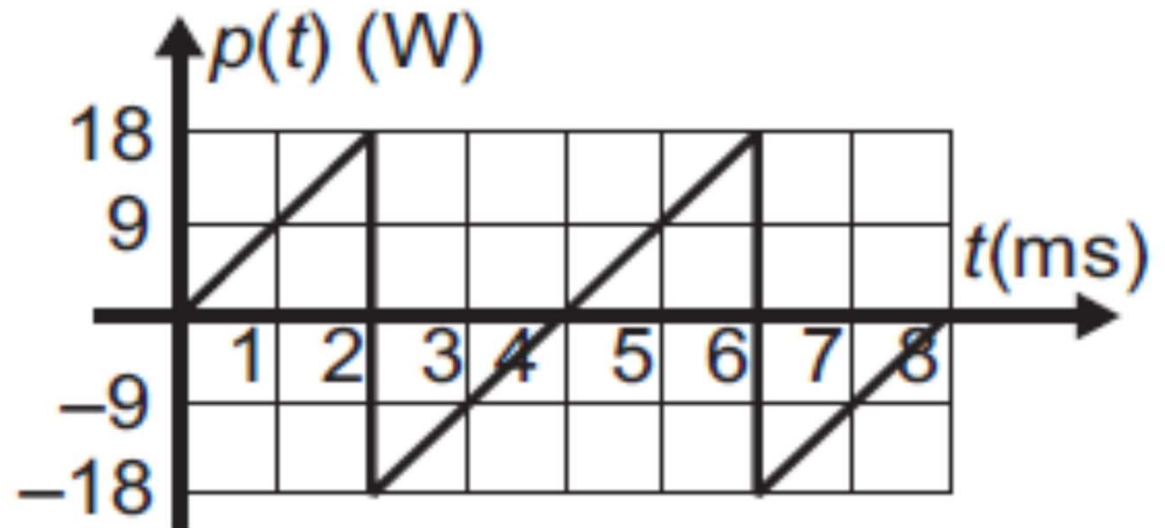
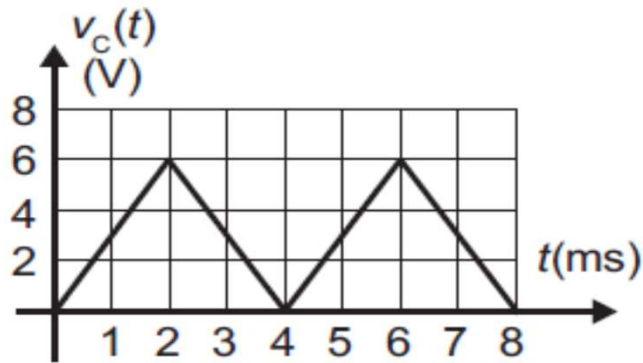
$$(6,8): v_c(t) = 24 - 3t$$

# Circuit Element: Capacitor



$$i_c(t) = C \frac{dv_c(t)}{dt}$$

# Circuit Element: Capacitor



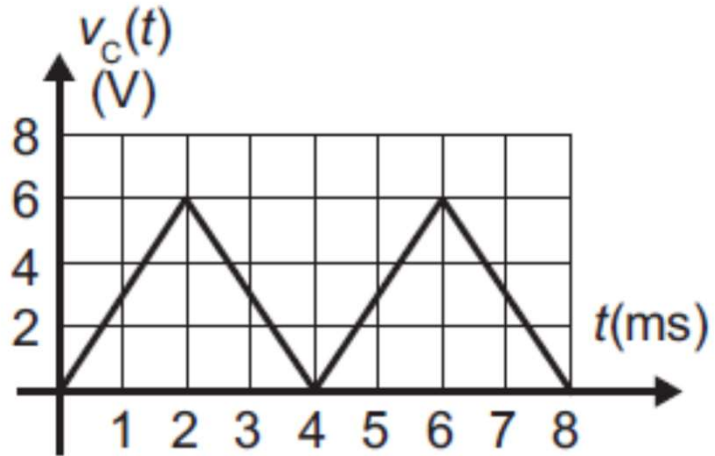
$$(0,2): P_L(t) = 9t$$

$$(2,4): P_L(t) = 9t - 36$$

$$(4,6): P_L(t) = 9t - 36$$

$$(6,8): P_L(t) = 9t - 72$$

# Circuit Element: Capacitor

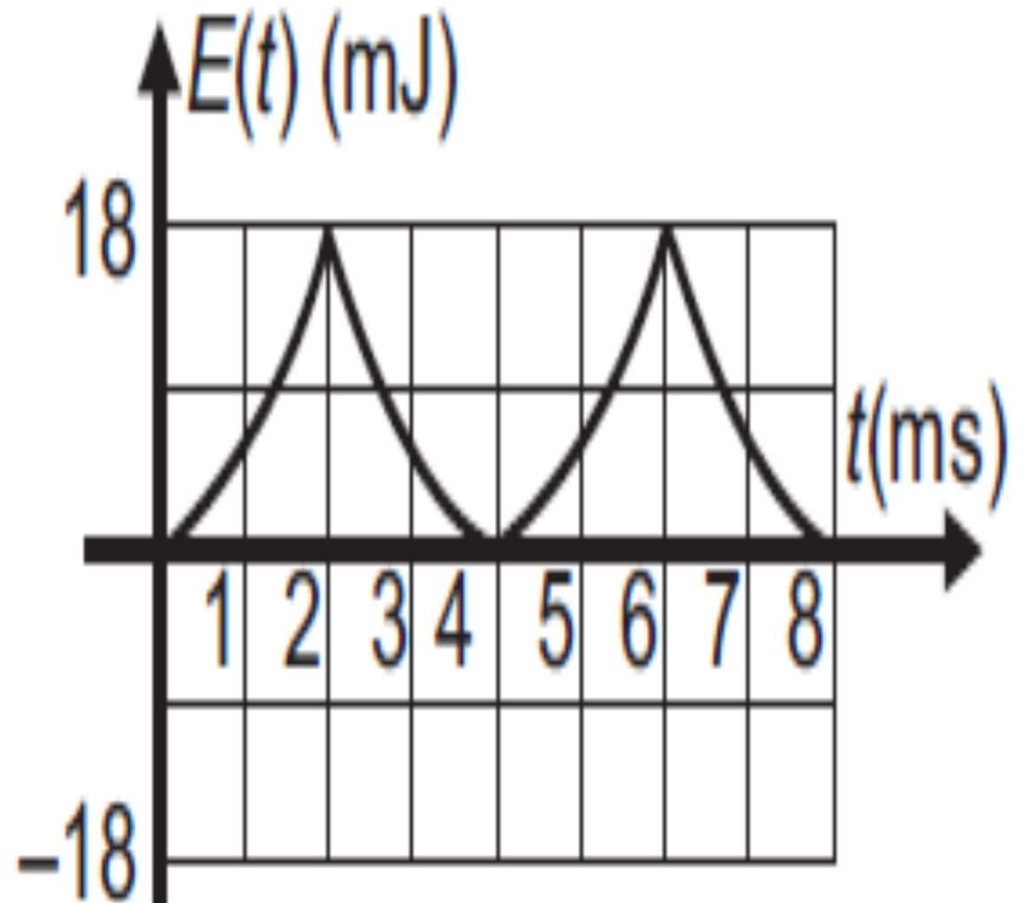


$$(0,2): E_L(t) = 4.5t^2$$

$$(2,4): E_L(t) = 4.5t^2 - 36t + 72$$

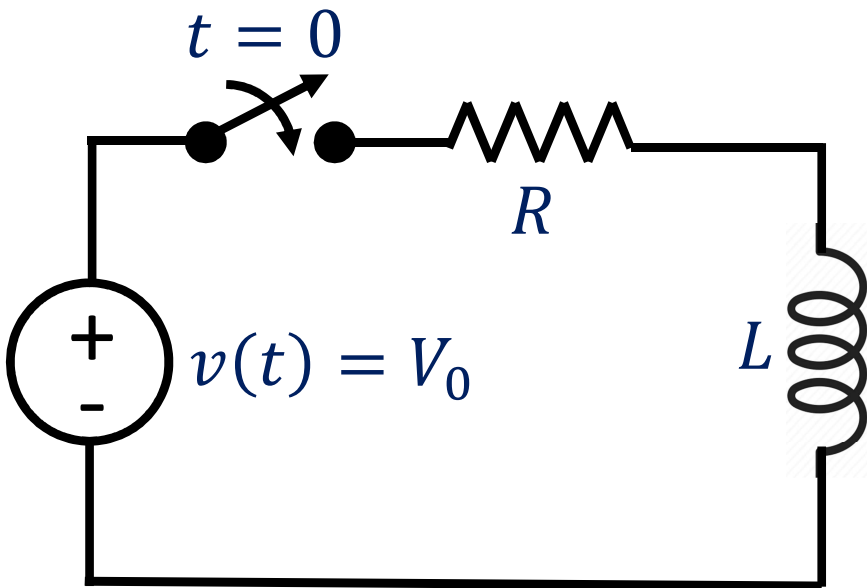
$$(4,6): E_L(t) = 4.5t^2 - 36t + 72$$

$$(6,8): E_L(t) = 4.5t^2 - 72t + 288$$





# RL Series Circuit

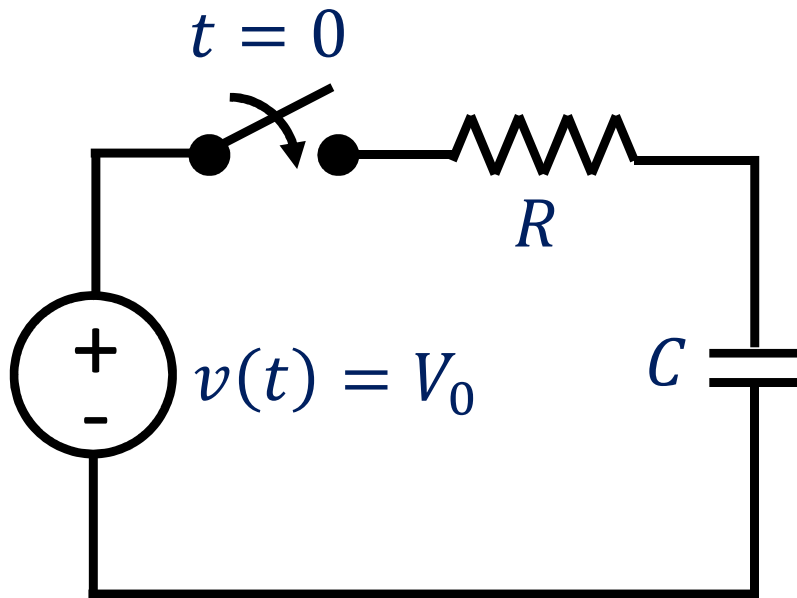


$$v(t) = Ri(t) + L \frac{di(t)}{dt} \quad (t > 0)$$

$$i(0) = i_L(0+) \quad (t = 0)$$

$$v(t) = V_0$$

# RC Series Circuit

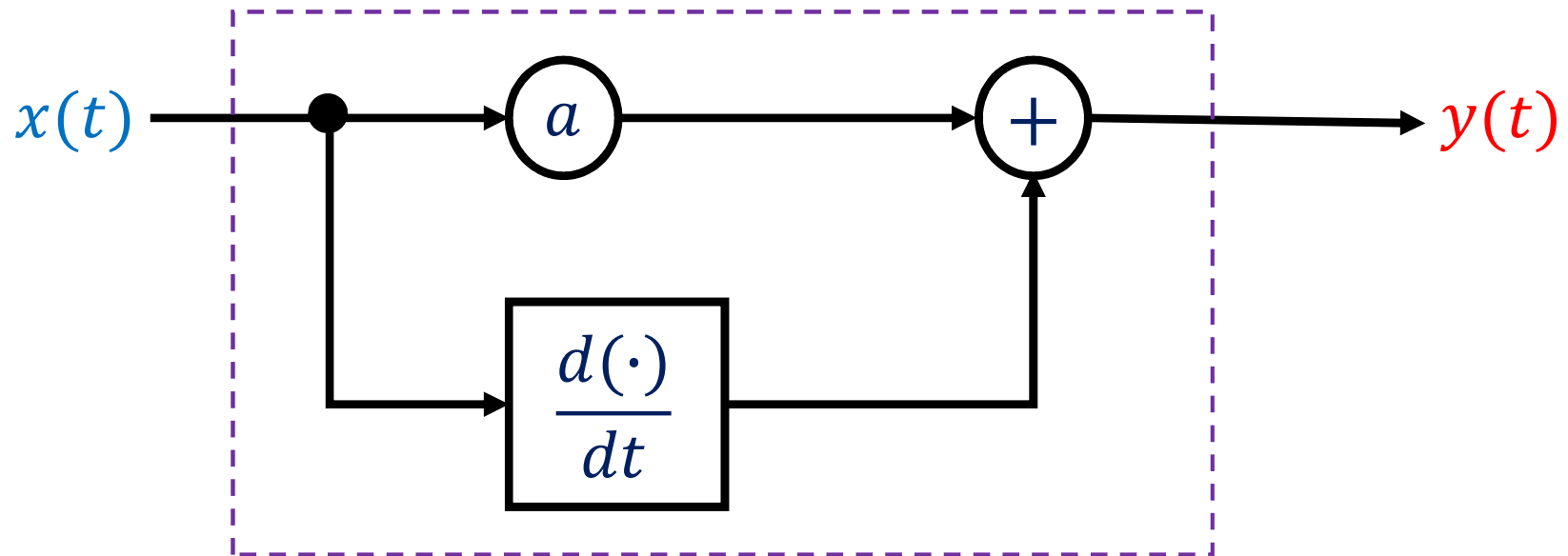


$$\left. \begin{aligned} \frac{v(t) - v_c(t)}{R} &= C \frac{dv_c(t)}{dt} \\ v(t) &= v_c(t) + RC \frac{dv_c(t)}{dt} \end{aligned} \right\} (t > 0)$$

$$t = 0 : v_c(0+)$$

# Response of a First Order Linear System

$$\frac{dx(t)}{dt} + ax(t) = y(t)$$



# Response of a First Order Linear System

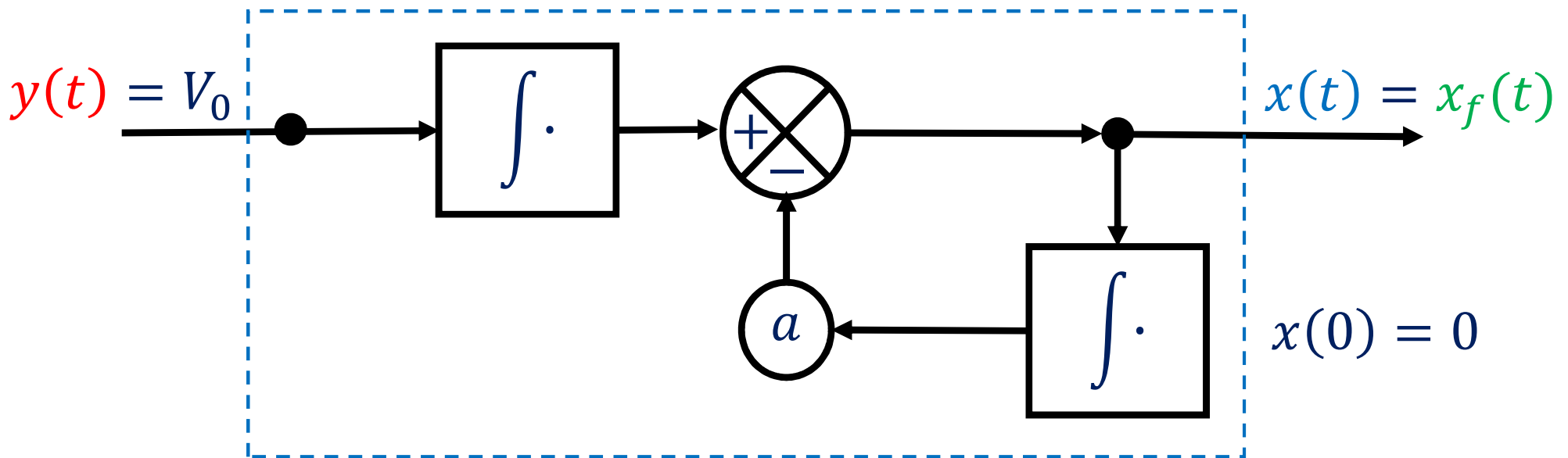
$$\frac{dx(t)}{dt} + ax(t) = y(t) \quad \longrightarrow \quad \begin{array}{l} \text{Solution: } x(t) = x_f(t) \\ \text{Particular Integral Solution OR} \\ \text{Forced Response} \end{array}$$

$$\frac{dx(t)}{dt} + ax(t) = 0 \quad \longrightarrow \quad \begin{array}{l} \text{Solution: } x(t) = x_n(t) \\ \text{Complementary Solution OR} \\ \text{Natural Response} \end{array}$$

$$\text{Complete Solution: } x(t) = x_f(t) + x_n(t)$$

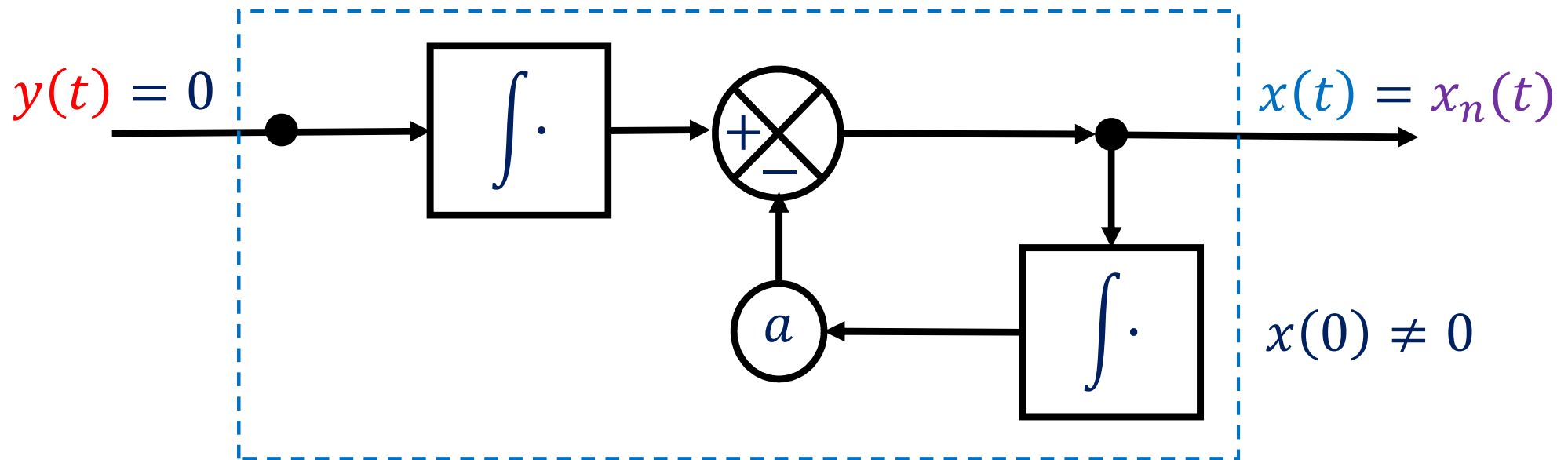
# Response of a First Order Linear System

$$\frac{dx(t)}{dt} + ax(t) = y(t) \quad \longrightarrow \quad x(t) = \int y(t)dt - a \int x(t)dt$$

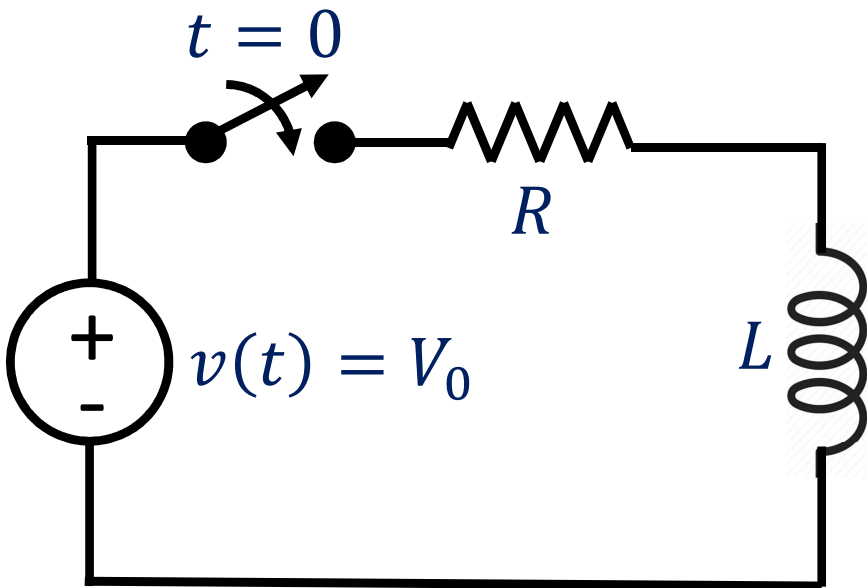


# Response of a First Order Linear System

$$\frac{dx(t)}{dt} + ax(t) = y(t) \quad \longrightarrow \quad x(t) = \int y(t)dt - a \int x(t)dt$$



# RL Series Circuit



$$V_0 = Ri(t) + L \frac{di(t)}{dt} \quad (t > 0)$$

$$i(0) = i_L(0+) \quad (t = 0)$$

## RL Series Circuit

$$Ri(t) + L \frac{di(t)}{dt} = V_0 \quad \Rightarrow \quad \int \frac{L d\{i(t)\}}{V_0 - Ri(t)} = \int dt$$

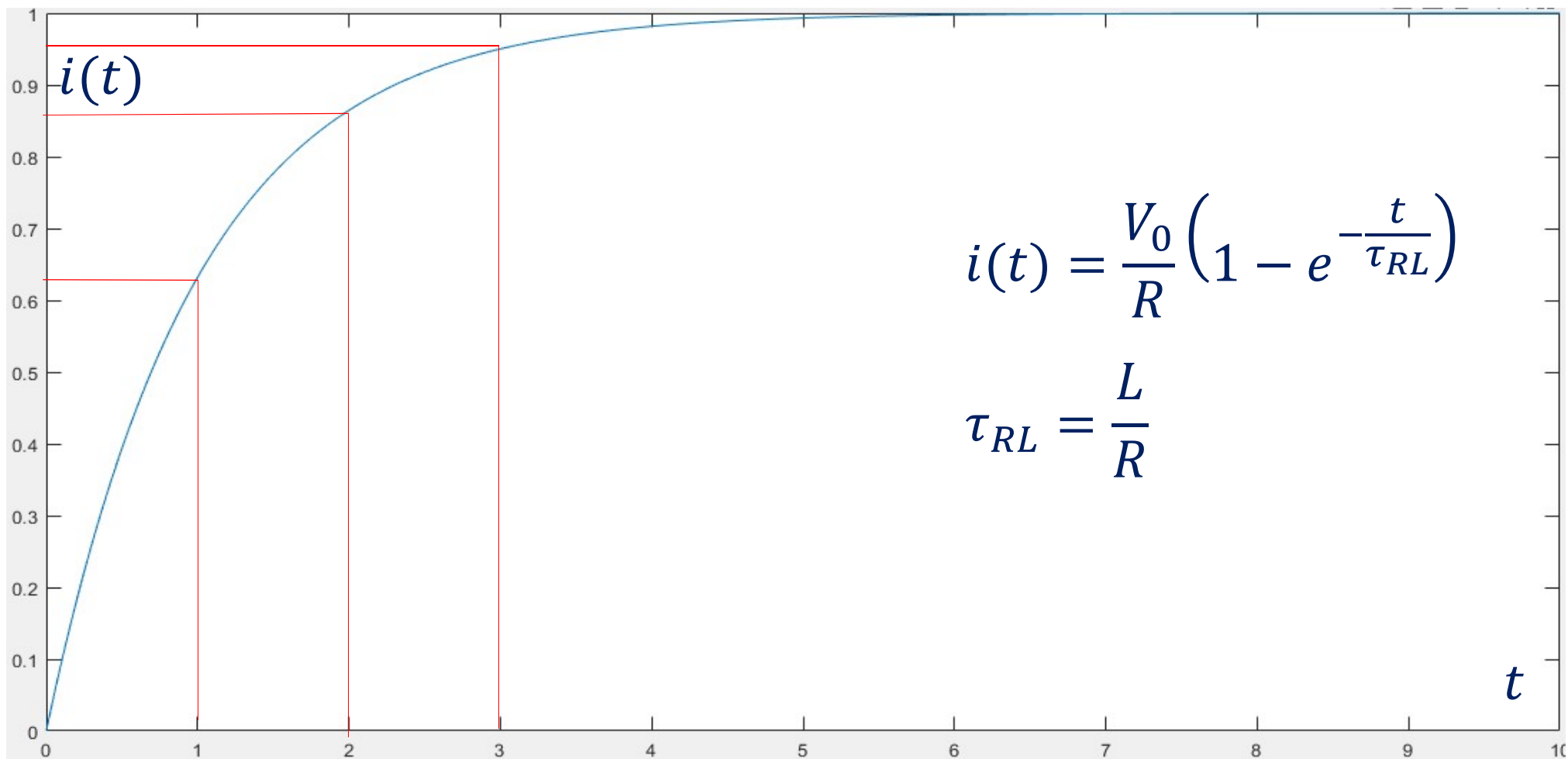
$$\Rightarrow -\frac{L}{R} \ln\{V_0 - Ri(t)\} = t + K$$


$$t = 0 \Rightarrow K = -\frac{L}{R} \ln\{V_0 - Ri_L(0+)\} \rightarrow \frac{L}{R} \ln \left\{ \frac{V_0 - Ri(t)}{V_0 - Ri_L(0+)} \right\} = -t$$

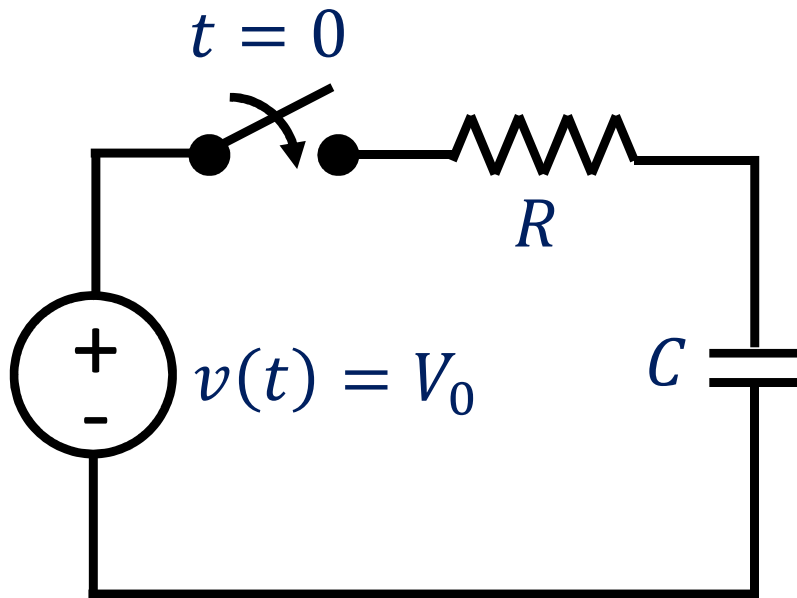
$$i(t) = \frac{V_0}{R} + \left\{ i_L(0+) - \frac{V_0}{R} \right\} e^{-\frac{R}{L}t} = \frac{V_0}{R} \{1 - e^{-\frac{R}{L}t}\} + i_L(0+) e^{-\frac{R}{L}t}$$



# RL Series Circuit



# RC Series Circuit



$$\left. \begin{aligned} \frac{V_0 - v_c(t)}{R} &= C \frac{dv_c(t)}{dt} \\ V_0 &= v_c(t) + RC \frac{dv_c(t)}{dt} \end{aligned} \right\} (t > 0)$$

$$t = 0 : v_c(0+)$$

## RC Series Circuit

$$V_0 = v_c(t) + RC \frac{dv_c(t)}{dt} \quad \Rightarrow \quad \int \frac{RC dv_c(t)}{V_0 - v_c(t)} = \int dt$$

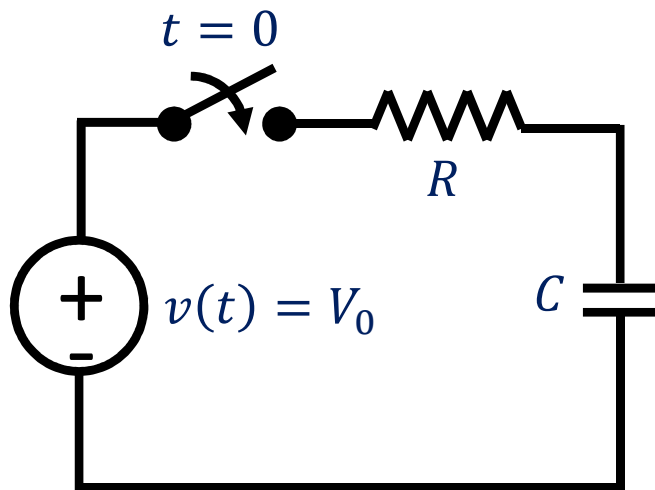
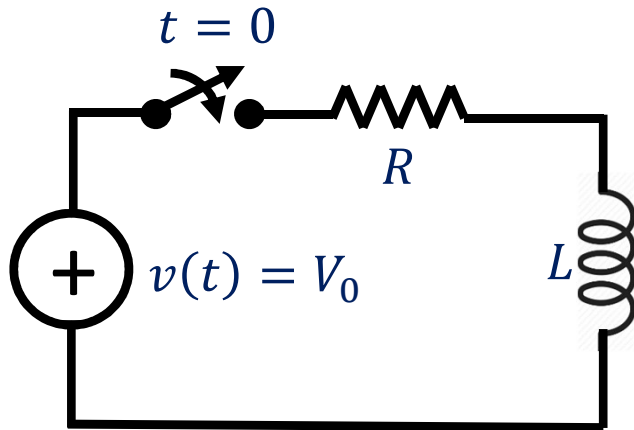
$$\Rightarrow -RC \ln\{V_0 - v_c(t)\} = t + K$$

↓

$$\underbrace{K = -RC \ln\{V_0 - v_c(0+)\}}_{t=0} \longrightarrow \ln \left\{ \frac{V_0 - v_c(t)}{V_0 - v_c(0+)} \right\} = -\frac{t}{RC}$$

$$v_c(t) = V_0 + \{v_c(0+) - V_0\}e^{-\frac{t}{RC}} = V_0\{1 - e^{-\frac{t}{RC}}\} + v_c(0+)e^{-\frac{t}{RC}}$$

# Response of First Order Linear Circuits



$$i_L(t) = \frac{V_0}{R} \{1 - e^{-\frac{R}{L}t}\} + i_L(0+)e^{-\frac{R}{L}t}$$

$$x(t) = x(\infty) \{1 - e^{-\frac{t}{\tau}}\} + x(0+)e^{-\frac{t}{\tau}}$$

$$v_c(t) = V_0 \{1 - e^{-\frac{t}{RC}}\} + v_c(0+)e^{-\frac{t}{RC}}$$

# Summary

- Introduced Inductor
  - Studied  $v_L(t)$ ,  $i_L(t)$ ,  $P_L(t)$ ,  $E_L(t)$
- Introduced Capacitor
  - Studied  $v_c(t)$ ,  $i_c(t)$ ,  $P_c(t)$ ,  $E_c(t)$
- First Order Linear System
  - Complete Solution:  $x(t) = x_f(t) + x_n(t)$
- RL Series Circuit
  - Forced and Natural Response:  $i_L(t)$
- RC Series Circuit
  - Forced and Natural Response:  $v_c(t)$



# Thank You